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MILITARY SPECIFICATION
ALTIMETER SET, ELECTRONIC AN/APN-209(V)
(UNITS OF)

This specification is approved for use within Communications-Electronics Command, Department of the Army, and is available for use by all Departments and agencies of the Department of Defense.

1. SCOPE

*1.1 SCOPE. This specification covers the following units of Radar Altimeters which provide an accurate indication of altitude of an aircraft over an altitude range of 0 to 1,500 feet and are designated Altimeter Set, Electronic AN/APN-209A(V), and AN/APN-209B(V) (See 6.1 and 6.4):

Antenna, AS-2595/APN-194(V)
Antenna, AS-3586/APN-209A(V)
Receiver-Transmitter, Height Indicator; RT-1115A/APN-209(V)
Receiver-Transmitter, Height Indicator; RT-1115B/APN-209(V)
Receiver-Transmitter, RT-1411/APN-209A(V)
Indicator, Height, ID-1917/APN-209(V)
Indicator, Height, ID-1917A/APN-209(V)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, U.S. Army Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Federal

L-P-378	Plastic Sheet Strip, Thin Gauge Polyolefin
NN-P-71	Pallet, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial)
QQ-S-781	Strapping, Steel and Seals
PPP-T-97	Tape, Packaging/Industrial, Filament Reinforced
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triplewall
PPP-B-1672	Box, Shipping, Reusable with Cushioning
PPP-S-760	Strapping, Nonmetallic (and Connectors)

Military

MIL-P-116	Preservation-Packing, Methods of
MIL-C-5541	Chemical Conversion Coatings and Aluminum and Aluminum Alloys
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-P-15024/6	Plate, Identification
MIL-L-25467	Lighting, Integral, Instrument, General Specification for
MIL-M-38510/104	Microcircuits, Linear, Line Drivers and Receivers, Monolithic Silicon
MIL-I-49077	Installation and Acceptance Testing of Altimeter Set, Electronic AN/APN-209 in Army Aircraft

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STANDARDS

Federal

FED-STD-595	Color

Military

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781B	Reliability Tests, Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1188	Commercial Packaging of Supplies and Equipment

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

*DRAWINGS

USAECOM (80063)

DL-SM-B-707893	Indicator, Height ID-1917/APN-209(V)
DL-SM-B-967122	Receiver-Transmitter, Height Indicator RT-1115A/APN-209(V)
DL-SM-B-967240	Indicator, Height, ID-1917A/APN-209(V)
DL-SM-B-967283	Receiver-Transmitter, Height Indicator RT-1115B/APN-209(V)
DL-SM-B-967310	Receiver-Transmitter RT-1411/APN-209A(V)

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DL-SM-B-967316	Antenna AS-2595/APN-194(V)
DL-SM-B-967321	Antenna AS-3586/APN-209A(V)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets of MS standards), the test of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regualations unless a specific exemption has been obtained.

3. REQUIREMENTS

*3.1 Description (See 4.18) - The Altimeter Set provides a continuous indication of absolute altitude of an aircraft above the surface of the earth and the features upon it by transmitting a radio frequency signal in the range of 4200 to 4400 MHz to the ground, receiving the reflected signal and indicates the altitude of the aircraft on the IRT unit and a remote indicator. The Altimeter set operates from an aircraft supply having a nominal voltage of 27.5 volts DC. Absolute altitude indications are accomplished by the use of the units listed in 1.1. The Receiver-Transmitter, Height Indicator (IRT) or Receiver-Transmitter (RT) transmits a radio frequency signal, receives the reflected signal, and derives the proper signals for 0 to 1500 feet radar altitude in accordance with the requirements of this specification. In addition, the Receiver-Transmitter, Height Indicator displays analog altitude, digital altitude and high and low altitude warnings. It provides these signals along with an auxiliary analog altitude signal, a reliability signal and a self test confidence signal. The Receiver-Transmitter also provides the auxiliary analog altitude, reliability and self test confidence signals. The Receiver-Transmitter, Height Indicator or Receiver-Transmitter shall be capable of driving a remote Height Indicator (RI).

3.2 First Article - The contractor shall furnish the specified number of First Article units on order, as required by the contract (see 4.3).

*3.3 Construction (see 4.11) - The unit(s) on order shall be constructed in accordance with the following:

DL-SM-B-707893	Indicator, Height ID-1917/APN-209(V)
DL-SM-B-967122	Receiver-Transmitter, Height Indicator RT-1115A/APN-209(V)
DL-SM-B-967240	Indicator, Height ID-1917A/APN-209(V)
DL-SM-B-967283	Receiver-Transmitter, Height Indicator RT-1115B/APN-209(V)

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DL-SM-B-967310	Receiver-Transmitter RT-1411/APN-209A(V)
DL-SM-B-967316	Antenna AS-2595/APN-194(V)
DL-SM-B-967321	Antenna AS-3586/APN-209A(V)

3.3.1 Parts, Materials and Processes (See 4.4) - In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirement therein, shall apply.

*3.3.2 Weight (See 4.14) - The weight of each completely assembled unit shall not exceed 4.2 pounds for the RT-1115( )/APN-209(V), 2.5 pounds for the ID-1917( )/APN-209(V), 0.65 pounds for AS-2595/APN-194(V), 3.4 pounds for the RT-1411( )/APN-209A(V), and 0.15 pounds for the AS-3586/APN-209A(V).

3.3.3 Controls (See 4.11) - All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

*3.3.4 Finish, Protective (See 4.4) - The units shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as screws, etc., and necessary touch-up after mounting. The final paint film on Type 1 surfaces shall be Film B in accordance with MIL-F-14072, except that the face of the AS-2595/APN-194(V) shall be in accordance with 3.10.1.9 and that of the AS-3586/APN-209A(V) shall be in accordance with 3.10.2.8.

3.4 Service Conditions (See 4.8) - The unit(s) on order shall meet the following service conditions:

3.4.1 Temperature-Altitude (See 4.8.1) - The unit(s) on order shall be subjected to and meet the temperature-altitude requirements when non-operating from -62°C through +85°C at sea level to 40,000 feet. The unit(s) shall be operable and conforming to limits specified from -54°C through +71°C at sea level to 30,000 feet with the exception that the accuracy may degrade to ±(6.5 feet + 3% of altitude) in the temperature ranges of -54°C to -40°C and +55°C to +71°C. Operation shall be intermittent over the temperature range of +55°C to +71°C. No damage shall result from exposure to temperatures in the range of -62°C to +85°C and pressures from sea level to 40,000 feet.

3.4.2 Humidity (See 4.8.2) - The unit(s) shall withstand up to 100 percent relative humidity including conditions wherein condensation takes place in the form of water. When exposed to the above conditions the unit shall suffer no damage in the non-operating condition and shall meet specified performance when operated.

3.4.3 Dust (See 4.8.3) - The unit(s) shall withstand exposure to dust particles as encountered in desert areas with winds up to 17 knots.

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3.4.4 Vibration (See 4.8.4) - The unit(s) normally mounted shall not suffer damage or fail to meet specified performance when subjected to vibration normally encountered in both airplanes and helicopters up to 500 Hz.

3.4.5 Shock (See 4.8.5)

3.4.5.1 Bench Handling (See 4.8.5.1) - The non-operating unit(s) shall be capable of withstanding the normal bench handling required during servicing.

3.4.5.2 Basic (See 4.8.5.2) - The unit(s) shall withstand the normal shocks encountered in airplanes and helicopters.

3.4.5.3 Crash Safety (See 4.8.5.3) - The normally mounted units shall show no failures of the attaching points when the unit(s) are subjected to the level of shock encountered in a crash situation. Bending and distortion is permitted.

3.4.6 Fungus (See 4.8.6) - The unit(s) shall show no evidence of viable fungus after being subjected to 28 days of fungus exposure.

3.4.7 Salt Fog (See 4.8.7) - The unit(s) shall be capable of withstanding exposure, in normal operating configuration, to salt fog for 48 hours and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or clogging or binding of moving parts.

3.4.8 Explosive Atmosphere (See 4.8.8) - When operated in an ambient explosive mixture of gas and air the units shall cause no ignition.

3.4.9 Acceleration (See 4.8.9) - The unit(s) shall be operable without degradation in specified performance, and shall sustain no physical damage, when exposed to steady-state acceleration levels up to 6 g for one (1) minute.

*3.4.10 Rain (See 4.8.10) -

(1) The antenna shall be operable without degradation in specified performance, and shall sustain no physical damage during periods of heavy precipitation with intermittent wind.

(2) The Receiver-Transmitter, Height Indicator RT-1115( )/APN-209(V) unit(s), Receiver-Transmitter RT-1411( )/APN-209A(V) and Indicator, Height ID-1917( )/APN-209(V) unit(s) shall withstand a "steady-state" rainfall on all surfaces, meet specified performance, and sustain no physical damage.

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*3.4.11 Leakage (Immersion) (see 4.8.11) - The Antennas AS-2595/APN-194(V) and AS-3586/APN-209A(V) shall be capable of immersion in water without leakage of water into the enclosure. (This requirement is only for the Antennas and does not apply to any other AN/APN- 209( )(V) equipment).

*3.4.12 Differential Pressure (See 4.8.12) - The Antennas AS-2595/APN-194(V) and AS-3586/APN-209A(V) shall be capable of withstanding a differential pressure across the radome of at least one atmosphere.

3.5 Numerical Reliability Requirement (See 4.6.2)

3.5.1 AN/APN-209( )(V)1 Altimeter Set - The equipment shall have a specified (θ_o) Mean-Time-Between-Failure (MTBF) of 1000 hours when operating in aggregate under the conditions of 4.6.2.2. This MTBF is allocated to system components as follows:

Receiver-Transmitter, Height Indicator	1,020 hours
Antenna	100,000 hours
Altimeter Set (θ_o)	1,000 hours

3.5.2 AN/APN-209( )(V)2 Altimeter Set - The equipment shall have a specified (θ_o) Mean-Time-Between-Failure (MTBF) of 800 hours when operating in aggregate under the conditions of 4.6.2.2. This MTBF is allocated to system components as follows:

Receiver-Transmitter, Height Indicator	1,020 hours
Height Indicator	4,000 hours
Antenna	100,000 hours
Altimeter Set (θ_o)	800 hours

*3.5.3 AN/APN-209( )(V)3 Altimeter Set - The equipment shall have a specified (θ_o) Mean-Time-Between-Failure (MTBF) of 2800 hours when operating in aggregate under the conditions of 4.6.2.2. This MTBF is allocated to system components as follows:

Receiver-Transmitter,	2,970 hours
Antenna	100,000 hours
Altimeter Set (θ_o)	2,800 hours

3.6 Electromagnetic Interference (See 4.10) - The emission of electromagnetic interference by the Altimeter Set and the susceptibility of the Altimeter Set to electromagnetic interference shall comply with the following requirements of MIL-STD-461 as specified herein.

3.6.1 Power Lead Conducted Emission (30 Hz to 50 KHz) Method CE01 - The narrowband conducted interference shall not exceed the limits shown in Figure 4 of MIL-STD-461, Notice 4.

*3.6.2 Signal Lead Conducted Emission (30 Hz to 50 KHz) Method CE03 - The narrowband conducted interference shall not exceed the limits shown on Figure 4 of MIL-STD-461, Notice 4. The test shall be conducted on the signal cable and shall include the following leads separated from their returns:

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(a) Self test confidence signal
(b) Self test input signal
(c) Auxiliary analog range signal
(d) 5 VAC light control
(e) Blanking pulse

3.6.3 Conducted Emission, Power and Signal Leads (50 KHz to 50 MHz) Methods CE04 and CE05 - The broadband conducted interference shall not exceed the limits shown on Figure 5 of MIL-STD-461, Notice 4 except as shown below and the narrowband conducted interference shall not exceed the limits shown on Figure 6 of MIL-STD-461, Notice 4.

MODIFIED CONDUCTED LIMITS - Test CE04 & CE05

Test CE04		Test CE05	
Freq	Max Limit	Freq	Max Limit
2-3 MHz	59 dB ua	1.5-6 MHz	56 dB ua
3-4 MHz	57 dB ua	6-10 MHz	59 dB ua
4-10 MHz	53 dB ua		

*3.6.4 Power Lead Conducted Susceptibility (30 Hz to 50 KHz) Method CS01 - The Altimeter and Indicator shall show no change in indication or performance greater than the specified tolerance when the 28 VDC power lines are subjected to audio signals at a level shown on Figure 8 of MIL-STD-461, Notice 4.

3.6.5 Power Lead Conducted RF Susceptibility (50 KHz to 400 MHz) Method CS02 - The Altimeter and Indicator shall show no change in indication or performance greater than the specified tolerance when the +28 V power line is subjected to RF signals at the level shown in Figure 19 of MIL-STD-461, Notice 4.

3.6.6 Conducted Susceptibility Intermodulation (430 MHz to 10 GHz) Method CS03 - The Altimeter analog range voltage shall not change by more than $\pm$((3 feet +.03 X altitude) X (7)) mV when performing the two signal intermodulation test.

3.6.7 Conducted Susceptibility, Rejection of Undesired Signals (3 KHz to 10 GHz) Method CS04 - The Altimeter analog range voltage shall not change by more than $\pm$((3 feet +.03 X altitude) X (7)) mV when performing the rejection of undesired signals test.

3.6.8 Conducted Susceptibility, Power Lead Spike Method CS06 - The Altimeter and Indicator shall show no change in indication or performance greater than the specified tolerance when the 28 V power lines are subjected to transients of 56 volts peak magnitude and 10 microseconds duration.

*3.6.9 Radiated Emission, Electric Field (14 KHz to 10 GHz) Method RE02 and RE02.1 - The broadband and narrowband radiated interference shall not exceed the limits shown in Figures 13 and 14 of MIL-STD-461, Notice 4, except as shown below. Emissions from the Receiver-Transmitter, Height Indicator or the Receiver-Transmitter at the operating frequency are excluded from the requirements of RE02.

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MODIFIED RADIATION EMISSION LIMITS - Test RE02

Broadband		Narrowband	
Freq	Max Limit	Freq	Max Limit
100-250 MHz	60 dB uV/M/MHz	22-28 MHz	31 dB uV/M/MHz

*3.6.10 Transmitter Spurious and Harmonic Emission Method RE03 - The spurious and harmonic radiated emission from the altimeter transmitting antenna shall be 70 dB below the level of the transmitter fundamental radiated emission (dBm/m^2) except as listed below:

4.085 - 4.515 GHz	No Limit
3.900 - 4.084 GHz	-20 dB
4.516 - 4.700 GHz	-20 dB
3.300 - 3.899 GHz	-40 dB
4.701 - 5.300 GHz	-40 dB
1.75 - 2.40 GHz	-55 dB

The spurious and harmonic radiated emission from the receiving antenna shall be less than -60 dBm except in the 4.29 GHz to 4.31 GHz frequency range. Emission in the 4.29 GHz to 4.31 GHz frequency range shall be no greater than -17 dBm.

3.6.11 Radiated Susceptibility, Magnetic Induction Field Spike Method RS02 - The Altimeter and Indicator shall show no change in indication or performance greater than the specified tolerance when 100 volts peak, 10 microsecond, spikes are applied to the radiating wire.

3.6.12 Electric Field Radiated Susceptibility (10 KHz to 12.4 GHz) Method RS03.2 - The Altimeter and Indicator shall show no change in indication or performance greater than the specified tolerance when the test setup is subjected to the electric fields listed below.

Frequency	Field Level
10 KHz - 2 MHz	1 V/M
2 MHz - 30 MHz	5 V/M
30 MHz - 200 MHz	10 V/M
200 MHz - 2 GHz	10 V/M
2 GHz - 12.4 GHz	5 V/M

3.7 Electrical Requirements (See 4.9) - Unless otherwise specified, the equipment shall be capable of meeting the following requirements.

3.7.1 Altitude Range (See 4.9.2) - The set shall accurately indicate absolute altitude from 0 to maximum range over all types of terrain surfaces, such as trees, foliage, deep snow, ice, land and water. The altimeter shall provide a digital output from 0 to 1500 feet, and an analog output from 0 to 1500 feet.

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3.7.2 Range Accuracy

3.7.2.1 Altitude Accuracy (See 4.9.3.1 and 4.18) - For altitudes up to 1500 feet, the output altitude signal levels, as measured on the digital portion of the indicators and the analog output signals, shall be within ±(3 feet + 3% of actual altitude) from the correct terrain clearance altitude.

3.7.2.2 Pointer Accuracy (See 4.9.3.1, 4.9.3.2 and 4.18) - The pointer of the indicators shall be within ± 2 feet or ±2% of the analog altitude voltage whichever is greater for altitude of 200 feet or less. Above 200 feet the pointer accuracy shall be ± 10 feet or 2% of the analog altitude voltage whichever is greater.

3.7.2.3 Remote Digital Altitude Display (See 4.9.3.2 and 4.18) - The remote indicator digital display shall be within ±1 foot of the digital output signal of the IRT or RT for altitudes below 255 feet, above 255 feet the remote digital altitude display shall be within 10 feet of the digital output signal of the IRT or RT.

3.7.2.4 Short Term Drift (See 4.9.21) - The short term drift at any constant temperature within the stated environmental requirements shall not exceed ±2 feet within 15 minutes as displayed on the digital portion of the indicators for altitude range between 0 and 200 feet.

3.7.3 Frequency (See 4.9.20) - The set shall operate within the allocated frequency range of 4200 to 4400 MHz with a center frequency of 4300 MHz.

3.7.4 Electrical Tracking Rate (See 4.9.16) - The altitude signal (Set less Indicator) shall follow changes in height at the rate of at least 1000 feet/second.

*3.7.5 System Sensitivity (see 4.9.4) - Minimum loop gain (transmit port to receive port) shall be 119 dB under standard conditions (0°C to $+40^{\circ}$C). At temperatures other than standard conditions the minimum loop gain shall be 117 dB. The transmitter peak power shall not exceed +38.5 dBm when measured at the transmitter port.

*3.7.6 Search Rate (See 4.9.17) - While in the search mode the system shall search from zero altitude to maximum at a rate of 4 ± 0.5 times per second.

3.7.7 Output Protection (See 4.9.14) - Protection circuitry shall be included in the system, both receivers and transmitters, to prevent damage due to accidental short-circuits or open-circuits of the electrical signal outputs (remote indicator outputs, auxiliary outputs, relay closures, etc.), and the RF signal outputs.

3.7.8 Multiple Indicator Protection (See 4.9.10) - When more than one indicator is used, isolation shall be provided between indicators to prevent a short-circuit to ground or incorrect voltage on one indicator from affecting other indicators.

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3.7.9 Installation Delay Adjustment (See 4.9.19) - The system shall be adjustable externally to allow for an aircraft installation delay from 1 to 70 feet. (Antenna height above ground and for RF coaxial cable lengths.)

3.7.10 Self Test Provisions (See 4.9.5) - A self test shall be incorporated. This shall be used for inflight and preflight testing of the set and shall designate failure at least down to the Line Replaceable Unit (LRU) level. Self test shall be activated by depressing the control knob on the Indicator and shall be capable of being energized continually with no degradation of performance effects to the set. Whenever self test is activated, the indicator display and electrical signal output shall indicate 1000 ± 100 feet. The self test function shall also be capable of being activated remotely by grounding pin 8 of J2 (auxiliary connector). A self test confidence logic output shall be provided by the altimeter at pin 13 of J2 (auxiliary connector) if the self test acceptance criteria have been met during the self test operation.

3.7.11 Input Electrical Power

*3.7.11.1 Operating Power (See 4.19) - Operating power shall be 27.5 VDC at 1.1 amperes maximum for the Receiver-Transmitter, Height Indicator, 27.5 VDC at 0.8 amperes maximum for the Receiver-Transmitter and 27.5 VDC at 0.6 amps maximum for the Indicator, Height. The equipment shall meet the performance as specified herein when energized from a power source having characteristics as defined in MIL-STD-704, Notice 2, for Category B utilization equipment. Steady-state DC voltage limits of operation shall be as specified in MIL-STD-704, Notice 2, Table II for normal, abnormal and landing modes of electric system operation. Reduced system performance is acceptable for emergency electrical system operation. If conflicts arise between the requirements of MIL-STD-704 and MIL-STD-461, Notice 4, the requirements of MIL-STD-461, Notice 4 shall apply.

3.7.11.2 Integral Lighting Power (See 4.9.15) - Input power for integral lighting shall require not more than 0.5 ampere at 5 VAC or VDC per indicator.

3.7.11.3 Dimming Control Power (See 4.9.15) - Input power for controlling the dimming of the digital display and warning lamps shall not exceed 6 mA at 27.5 VDC per indicator.

*3.7.11.4 Power On Control (See 4.20) - Each unit shall be turned on when the LO set knob on the indicator is rotated to position the LO warning index at or above zero feet. Additionally, each unit shall switch input power to a separate connector pin for use as a system power on control. The unit shall be turned on whenever the signal is +25 ± 5 VDC even though the power control knob on the indicator is off. The Receiver-Transmitter shall be turned on when 27.5 VDC power is applied to the power input at J1 pin 9 and the signal on the power on control line J2 pin 5 is +25 ± 5 VDC.

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3.7.11.5 Transient Spike Voltages (see 4.19.1) - The altimeter shall show no degradation in performance after being subjected to a transient spike voltage of ± 600 volts in magnitude and 10 microseconds duration; decreasing linearly from + 600V to +200V and -600V to -170V, in 40 microseconds; decreasing linearly from +200V to +70 and -170V to +7V in 400 microseconds; and remaining at these levels for 9.5 milliseconds. This spike shall be cycled at least 60 times within one hour.

*3.7.12 Altitude Memory (See 4.9.18) - All "in track" outputs of the Altimeter shall be maintained for 1.8 ± 0.5 seconds after loss of return signal before reverting to the search (out of track) state.

*3.7.13 Data Lag (See 4.9.22) - Electrical analog data from the Receiver-Transmitter, Height Indicator or the Receiver-Transmitter shall not lag any simulated or actual altitude by more than 0.1 second when tracking at a rate up to 1000 feet per second. This requirement does not pertain to the digital altitude signal inasmuch as the digital data is averaged over a period of 0.44 second.

*3.7.14 Dimming (See 4.9.8.3) - Dimming of the digital display and warning lamps shall be controlled separately from the dial lighting. The dimming voltage shall be 0 to 27.5 VDC. The digital display shall be dimmable from full brightness with 27.5 VDC input to full off with 5 VDC or less input. The warning lamps shall be dimmable from full brightness with 27.5 VDC input to a level visible only with night vision devices such as AN/PVS-5. Maximum current required by the dimming circuits shall not exceed 6 milliamperes at 27.5 VDC. An AN/PVS-5 will be furnished as government loaned property.

3.7.15 Integral Lighting (See 4.9.23) - The indicators shall be red internal lighted in accordance with MIL-L-25467.

*3.7.16 Zero Altitude Power (see 4.9.27) - The IRT or RT shall have a peak output power level of +27 dBm at an altitude of zero feet.

3.8 Electrical Signal Output Characteristics

3.8.1 Altitude Outputs (See 4.9.12) - The system shall provide sufficient isolated analog DC voltages proportional to altitude to drive its internal indicator and remote indicator. This signal shall be 0 to -10.5 VDC representing 0 to 1500 feet respectively. During no-track or unreliable operation, this signal shall be -13 ± 2 VDC.

3.8.2 Auxiliary Altitude Outputs (See 4.9.6) - In addition to the outputs specified in 3.8.1, the system shall provide a separate isolated linear DC voltage proportional to altitude, having a fixed scale factor of -7 millivolts per foot. During search or unreliable conditions this signal shall be +0.75 ± 0.5 VDC and shall be capable of driving a 2000 ohms or greater resistive load.

3.8.3 Reliability Signal (See 4.9.7) - The system shall provide a 4.5 ± 1.0 VDC reliability signal voltage during track or reliable operation and 0.5 ± 0.5 VDC during search or unreliable system operation. This signal shall be TTL compatible.

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3.8.4 High Altitude and Low Altitude Trip Tests (See 4.9.9) - The LO warning lamp shall be on when the indicated digital altitude is below the LO set index and off when the indicated digital altitude is above the LO set index. When descending between 1500 feet and 200 feet, activation of the LO warning shall occur when the indicated altitude is within +50, -20 feet, or +5, -2 percent (whichever is greater) of the selected warning altitude. When descending below 200 feet, actuation of the warning shall occur when the indicated altitude is within +5, -2 feet or +5, -2 percent (whichever is greater) of the selected warning altitude.

Below 200 feet when ascending, deactivation of the warning shall occur when the indicated digital altitude is within 7 feet or 5 percent, whichever is greater, above the point of activation of the warning. Above 200 feet when ascending, deactivation of the warning shall occur when the indicated digital altitude is within 70 feet or 5 percent, whichever is greater, above the point of activation of the warning.

The HI warning lamp shall be on when the indicated digital altitude is above the HI set index and off when the indicated digital altitude is below the HI set index. When ascending between 0 and 200 feet, activation of the warning shall occur when the indicated digital altitude is within +2, -5 feet or +2, -5 percent (whichever is greater) of the selected warning altitude. When ascending above 200 feet, actuation of the warning shall occur when the indicated digital altitude is within +20, -50 feet or +2, -5 percent (whichever is greater) of the selected warning altitude. Below 200 feet, when descending, deactivation of the warning shall occur when the indicated digital altitude is within 7 feet or 5 percent, whichever is greater, below the point of activation of the warning. Above 200 feet when descending, deactivation of the warning shall occur when the indicated digital altitude is within 70 feet or 5 percent, whichever is greater, below the point of activation of the warning.

3.8.4.1 Low Altitude Warning Signal (See 4.9.9) - The low altitude warning function shall be activated when the actual altitude displayed is at or below the pilot adjustable low altitude limit setting. The output shall be controlled by a transistor switch. The transistor shall be on when the warning is activated and off when there is no warning. When on the output voltage shall be 1.2 V maximum, when sinking 80 milliamperes. When off, the output shall be an open collector and capable of withstanding 50 VDC.

3.8.4.2 High Altitude Warning Signal (See 4.9.9) - The high altitude warning function shall be activated when the actual altitude displayed is at or above the pilot adjustable high altitude limit setting. The output shall be controlled by a transistor switch. The transistor shall be on when the warning is activated and off when there is no warning. When on the output voltage shall be 1.2 V maximum, when sinking 80 milliamperes. When off, the output shall be an open collector and capable of withstanding 50 VDC.

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3.8.5 Push-To-Test (PTT) Control (See 4.9.13) - The Height Indicator shall provide a ground to initiate system self test. This signal shall be open circuit at all other times.

*3.8.6 Remote Digital Altitude Signal (See 4.9.24) - The Receiver-Transmitter, Height Indicator and Receiver-Transmitter shall provide a digital altitude signal for use in a remote indicator. This signal shall be a series of pulses with each pulse equal to 1 foot of altitude. Pulses shall be transmitted from a differential line driver in accordance with MIL-M-38510/104. The signal shall have the following characteristics:

(a)	Pulse Width:	100 nanoseconds minimum
(b)	Rise Time:	50 nanoseconds maximum
(c)	Frequency:	150 KHz maximum
(d)	Amplitude:	4.5 ± 1.0 volts peak

*3.8.7 Remote Enable Pulse (See 4.9.25) - The Receiver-Transmitter, Height Indicator and Receiver-Transmitter shall provide an enable pulse during the time that the digital altitude pulses are being transmitted. The pulse shall be transmitted from a differential line driver in accordance with MIL-M-38510/104 and shall have the following characteristics (See 6.7):

(a)	Lead Time:	15 microseconds minimum
(b)	Lag Time:	75 microseconds maximum
(c)	Rise Time:	50 nanoseconds maximum
(d)	Width:	0.20 ± 0.06 seconds
(e)	Frequency:	2.7 ± 0.8 Hz
(f)	Amplitude:	4.5 ± 1.0 volts

*3.8.8 Blanking Pulse (See 4.9.26) - The Receiver-Transmitter, Height Indicator and Receiver-Transmitter shall provide a blanking pulse during the time that the transmitted RF pulse from J4 is being transmitted. The blanking pulse shall have the following characteristics:

(a)	Width:	2.6 ± 0.3 microseconds
(b)	Amplitude:	6.0 ± 0.75 volts
(c)	PRF:	18 + 4, -6 KHz
(d)	Source Impedance On (+6V):	15 ohms maximum to +6V
(e)	Source Impedance Off (0V):	1050 ohms to ground maximum
(f)	Lead Time:	2.0 ± 0.2 microseconds

*3.9 Display Functions (See 4.9.8) - The difference between the AN/APN-209A(V) and AN/APN-209B(V) involves the indicator display features. The AN/APN-209A(V) has red digital displays and amber warning lights. The AN/APN-209B(V) has green digital displays and green infrared filtered warning lamps. See paragraph 6.1 for specific system configurations. The RT-1411/APN-209A(V) Receiver-Transmitter does not contain display functions.

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*3.9.1 RT-1115( )/APN-209(V) (See 4.9.8.1) - Digital altitude shall be indicated on a four digit lighted readout. Above 255 feet, the digital altitude shall be rounded to the nearest 10 feet. Analog altitude shall be indicated from 0 to 1500 feet by means of a single pointer against a fixed dial. Movable markers positioned against the same dial shall indicate the pre-selected height above or below which the high and low altitude warnings operate. Unreliable operation shall be indicated by an "OFF" flag coming into view, the pointer shall go behind the masked portion of the Indicator dial and the digital readout and the high and low warning lamps blanked when:

(a) The signal strength to the Receiver-Transmitter, Height Indicator becomes inadequate to provide reliable altitude information (See 3.7.5) or

(b) Aircraft altitude is between 1500 and 2000 feet. (Unreliable indications shall remain in view at altitudes above 2000 feet.)

3.9.1.1 Power Failure and Power Off Display Functions (See 4.9.11) - The unreliable operation indications of 3.9.1 shall occur when 28 VDC power to the system is lost, except that the pointer need not go behind the masked portion of the indicator dial.

3.9.2 ID-1917( )/APN-209(V) (See 4.9.8.2) - The Height Indicator shall provide remote display of the analog and digital altitude signals generated by the Receiver-Transmitter, Height Indicator or Receiver-Transmitter. The display features shall be identical to those provided by the Receiver-Transmitter, Height Indicator.

3.10 Antennas AS-2595/APN-194(V) and AS-3586/APN-209A(V)

3.10.1 Antennas AS-2595/APN-194(V) - The antenna shall be NSN 5841-100-181-0330 and shall be in accordance with SM-D-967316.

*3.10.1.1 Impedance - (For reference only) The nominal antenna impedance shall be 50 ohms.

*3.10.1.2 Beam Width (See 4.17.1.1) - The antenna shall provide a 3 dB, one-way beam width of 45 ± 5 degrees in the H plane by 45 ± 5 degrees in the E plane.

*3.10.1.3 Antenna Gain (See 4.17.1.2) - The antenna one-way gain shall be not less than 10.5 dB.

*3.10.1.4 V.S.W.R (See 4.17.1.3) - The free space VSWR of the antenna and connector shall be not greater than 1.3 at 4.30 GHz and 1.4 at 4.25 and 4.35 GHz.

*3.10.1.5 Antenna Isolation (See 4.17.1.4) - The antenna electrical isolation shall be not less than 85 dB at a spacing of two feet, center to center when flush mounted on a ground plane. The isolation test shall be made at 4.30 GHz.

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*3.10.1.6 Antenna Pattern (See 4.17.1.5) - The main lobe shall be in accordance with 3.10.1.2. Pattern shape tests shall be made at 4.30 ± .005 GHz. The one-way gain 22.5 + 2.5 degrees either side of boresight shall be not less than 7.5 dB.

*3.10.1.7 Time Delay of RF Signal - (For reference only) The variation in delay over all environments shall not exceed 0.1 nanoseconds. The total delay from 4.25 to 4.35 GHz shall not exceed one nanosecond.

3.10.1.8 Signal Rejection (See 4.17.1.6) - The signal rejection from the input to antenna output shall be not less than the following:

(a) 25 dB at 8.35 and 8.85 GHz.
(b) 50 dB from 8.50 to 8.70 GHz.

*3.10.1.9 Antenna Finish (See 4.4) - The surface treatment of exposed aluminum shall be chemical film per MIL-C-5541 or cadmium plate followed by clear irridite or dichromate and finished with one coat epoxy paint primer and one coat gray epoxy paint, color 36440 per FED-STD-595. Both sides of the mounting flange shall be free of paint, except if radome material encompasses the entire flange area only the opposite side of the flange need be clear of paint. The radome shall be free of paint except for marking in accordance with 3.10.1.10.

*3.10.1.10 Radome Marking (See 4.17.1.7) - The radome shall be marked "DO NOT PAINT" in 0.5 ± 0.1 inch characters, "H PLANE" in 0.25 ± .05 inch characters and double arrow using gray epoxy paint, color 36440 per FED-STD-595.

*3.10.2 Antenna AS-3586/APN-209A(V) - The antenna shall be in accordance with SM-D-967321.

*3.10.2.1 Impedance - (For reference only) The nominal antenna impedance shall be 50 ohms.

*3.10.2.2 Beamwidth (See 4.17.2.1) - The antenna shall provide a 3 dB, one-way beamwidth of 30 degrees full angle minimum in the H-plane and 30 degrees full angle minimum in the E plane.

*3.10.2.3 Gain (See 4.17.2.2) - The antenna one-way gain shall be 9 dB minimum.

*3.10.2.4 V.S.W.R. (See 4.17.2.3) - The free space VSWR of the antenna shall be 1.8 maximum at 4300 MHz, 2.2 maximum from 4275 to 4325 MHz and 3.0 maximum from 4250 to 4275 MHz and from 4325 to 4350 MHz.

*3.10.2.5 Isolation (See 4.17.2.4) - The antenna electrical isolation shall be 85 dB minimum at a spacing of 2 feet, center to center spacing when flush mounted on a ground plane. The isolation test shall be made at 4.30 GHz.

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*3.10.2.6 Antenna Pattern (See 4.17.2.5) - The main lobe shall be shaped to provide the beamwidth described in 3.10.2.2. The one way gain 15.0 ± 2.5 degrees either side of boresight shall be 6.0 dB minimum. Pattern shape tests shall be conducted at 4.30 ± 0.005 GHz.

*3.10.2.7 Time Delay of RF Signal - (For reference only) The variation in delay over all environments shall not exceed 0.1 nanoseconds. The total delay from 4.25 to 4.35 GHz shall not exceed five nanoseconds.

*3.10.2.8 Antenna Finish - The back surface shall be tin plated. The radome surface shall be unpainted.

3.11 Burn-In/Debugging and Preconditioning (See 4.6.1)

3.11.1 Burn-In/Debugging (See 4.6.1.1) - Each unit on order shall meet the requirements herein after subjection to the Burn-in/Debugging as specified in 4.6.1.1. Antennas are excluded from the burn-in requirement.

*3.11.2 Pre-Conditioning (See 4.6.1.2) - Prior to Burn-In/Debugging, the Receiver-Transmitter, Height Indicator and Receiver-Transmitter shall be subjected to the preconditioning specified in 4.6.1.2.

3.12 Interchangeability (See 4.12) - Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of military standard MIL-STD-454 and shall be physically and functionally interchangeable. Items which are covered by specified electrical performance requirements shall meet such specified performance requirements prior to installation into the unit(s). After installation of these items and prior to submission of unit(s) for the quality performance specifications, adjustments shall be made, if required, to permit the unit(s) to meet full equipment specification requirements. Alterations of any items, or the unit(s) shall not be permitted.

3.13 Operational Inspection (See 4.13) - Immediately prior to packaging and after applicable processes, inspections and tests are completed, each unit(s) shall be subjected to the operational inspection as specified.

3.14 Marking (See 4.4) - Markings shall be in accordance with MIL-M-13231 except that the nameplates shall be in accordance with Figure 1 of MIL-P-15024/6.

3.15 Workmanship (See 4.11) - The equipment shall be manufactured and assembled in accordance with the equipment drawings (see 3.3) and the following requirements:

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MIL-STD-454

Soldering	Requirement 5
Workmanship	Requirement 9

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility For Inspection - Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of Inspection - Inspection shall be classified as follows:

(a) First article inspection (does not include packaging). (See 4.3.)

(b) Inspection covered by subsidiary documents. (See 4.4.)

(c) Quality conformance inspection of equipment before packaging. (See 4.5.)

(d) Quality conformance inspection of packaging. (See 4.15.)

4.3 First Article - Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor. (See 3.2)

4.3.1 First Article Units - The contractor shall furnish that number of first article units as specified in the contract or purchase order.

4.3.2 First Article Inspection - The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, Burn-In, and the inspections specified for Group A, Group B, Group C and Group D (See Table I, II, III and IV respectively). The inspection shall be performed in the following order: 4.4, Burn-In, Group A and Group B for all units, Group D, and designated units for specific subgroups within Group C in accordance with 4.5.3.1. After completion of Group C and Group D inspection, conforming units shall be re-inspected and shall pass all Group A inspection. The Group D inspection may be conducted concurrently with Group C inspections.

4.3.3 First Article Data - First article test plan and test report(s) shall be as required in the contract.

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4.4 Inspection Covered By Subsidiary Documents - The following shall be inspected under the applicable subsidiary documents, as part of the inspection before preparation for delivery:

Inspections	Reqt Para
Parts, Materials and Processes	3.3.1
Protective Finish	3.3.4
Antenna	3.10.1.9, 3.10.2.8
Marking	3.14

4.5 Quality Conformance Inspection of Equipment Before Packaging - Following the reliability screening each unit shall be subjected to Burn-In/Debugging (see 3.11.1) prior to conformance inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement after Burn-In/Debugging shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor at the contractor's facility.

4.5.1 Group A Inspection - Each unit on contract shall be inspected for conformance to the inspections specified in Table I. Discrete lots shall be formed on all units that pass this inspection. Factors of lot composition not defined herein, or in the contract, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105; using the general inspection levels and AQL's indicated in Table I.

4.5.1.1 Order of Inspection Within Group A - Group A inspection shall be performed in an order satisfactory to the Government except that Operational shall be last and performed immediately prior to packaging.

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*TABLE I - Group A Inspection

INSPECTION	REQT. PARA.	INSP. PARA.	AQL MAJOR	AQL MINOR
Visual and Mechanical	3.15,3.3 3.3.3 3.10.1.10	4.11, 4.17.1.7		
Modules:			0.65%	2.5%
Assemblies:			0.65%	2.5%
RT-1115( )/APN-209(V) and RT-1411/APN-209A(V) less case			0.65%	2.5%
RT-1115( )/APN-209(V) and RT-1411/APN-209A(V) assembled			1.0%	4.0%
ID-1917( )/APN-209(V) less case			0.65%	2.5%
ID-1917( )/APN-209(V) assembled			1.0%	4.0%
AS-2595/APN-194(V) and AS-3586/APN-209A(V) assembled			1.0%	4.0%
Electrical				**
RT-1115( )/APN-209(V) and RT-1411/APN-209A(V):				
Altitude range***	3.7.1	4.9.2.1		
Altitude accuracy	3.7.2.1	4.9.3.1		
System sensitivity	3.7.5	4.9.4		
Self test	3.7.10	4.9.5.1, 4.9.5.3	1.0%	
Auxiliary alt output	3.8.2	4.9.6		
Reliability signal	3.8.3	4.9.7	Combined	
Display, function***	3.9.1	4.9.8.1		
Operational inspection	3.13	4.13		
ID-1917( )/APN-209(V):				
Display, function	3.9.2	4.9.8.2		
Self test	3.7.10	4.9.5.2	1.0%	
Altitude range	3.7.1	4.9.2.2	Combined	
Altitude accuracy	3.7.2	4.9.3.2		
Operational inspection	3.13	4.13		

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*TABLE I - Group A Inspection (Continued)

INSPECTION	REQT. PARA.	INSP. PARA.	AQL	
			MAJOR	MINOR
AS-2595/APN-194(V) and AS-3586/APN-209A(V):				
VSWR Test (AS-2595)	3.10.1.4	4.17.1.3	1.0%	
(AS-3586)	3.10.2.4	4.17.2.3		
Operational inspection	3.13	4.13		

**All electrical defects are considered major.
***Not applicable to the RT-1411/APN-209A(V).

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4.5.2 Group B Inspection - Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table II and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B Sampling Plans - The Group B AQL shall be 6.5% and the inspection level shall be S4.

4.5.2.2 Order of Inspection Within Group B - Group B inspection shall be performed in any order which is satisfactory to the Government.

4.5.3 Group C Inspection - This inspection shall be as listed in Table III and shall be performed on sample unit(s) that have been subjected to and met the Group A and Group B inspections. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.3.2.

4.5.3.1 Order of Inspection Within Group C - Group C inspections shall be performed in the order shown in each subgroup in Table III. Subgroup I tests shall be divided between the two units selected for the test in any manner acceptable to the procuring agency. Subgroup II and subgroup III tests shall be divided between the two units selected for test in accordance with the recommended test sequence in MIL-STD-810.

4.5.3.2 Sampling for Group C Inspection - Units shall be selected at random for each group of inspections listed in Table III as specified in 4.5.3.2.1 through 4.5.3.2.4. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot(s).

4.5.3.2.1 Subgroup I - For this subgroup, two units from the first forty (40) production units shall be selected for the first Group C inspection. For subsequent inspection, two units per month shall be selected.

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*TABLE II - Group B Inspection

INSPECTION	REQT. PARA.	INSP. PARA.	AQL
Electrical:			
RT-1115( )/APN-209(V) and RT-1411/APN-209A(V)			
High Altitude and Low Altitude Trip Tests**	3.8.4	4.9.9.1	6.5%
Power Failure Display**	3.9.1.1	4.9.11	Combined
Altitude Outputs	3.8.1	4.9.12	
Interchangeability	3.12	4.12	6.5%
ID-1917( )/APN209(V):			
High Altitude and Low Altitude Trip Tests	3.8.4	4.9.9.2	6.5%
Interchangeability	3.12	4.12	6.5%

**Not applicable to the RT-1411/APN-209A(V).

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*TABLE III - Group C Inspection

INSPECTION	REQT. PARA.	INSP. PARA.
Subgroup I: (RT-1115( )/APN-209(V), RT-1411( )/APN-209A(V) and ID-1917( )/APN-209(V) only)		
Push-To-Test Controls	3.8.5	4.9.13
Output Protection	3.7.7	4.9.14
Input Electrical Power	3.7.11.2, 3.7.11.3	4.9.15
Multiple Indicator Protection	3.7.8	4.9.10
Electrical Tracking Rate	3.7.4	4.9.16
Search Rate	3.7.6	4.9.17
Altitude Memory	3.7.12	4.9.18
Installation Delay	3.7.9	4.9.19
Frequency	3.7.3	4.9.20
Short Term Drift	3.7.2.4	4.9.21
Data Lag	3.7.13	4.9.22
Display Function (Dimming)	3.7.14	4.9.8.3
Remote Digital Altitude Signal	3.8.6	4.9.24
Remote Enable Pulse	3.8.7	4.9.25
Power On Control	3.7.11.4	4.20
Integral Lighting	3.7.15	4.9.23
Compatibility	3.1, 3.7.2	4.18
Zero altitude power	3.7.16	4.9.27
Blanking pulse	3.8.8	4.9.26
Subgroup II:		
Temperature-Altitude	3.4.1	4.8.1
Humidity	3.4.2	4.8.2
Vibration	3.4.4	4.8.4
Subgroup III:		
Weight	3.3.2	4.14
Dust	3.4.3	4.8.3
Bench Handling Shock	3.4.5.1	4.8.5.1
Basic Shock	3.4.5.2	4.8.5.2
Crash Safety Shock	3.4.5.3	4.8.5.3
Fungus	3.4.6	4.8.6
Salt Fog	3.4.7	4.8.7
Explosion	3.4.8	4.8.8
Acceleration	3.4.9	4.8.9
Rain	3.4.10	4.8.10
Leakage	3.4.11	4.8.11
Differential Pressure	3.4.12	4.8.12

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*TABLE III - Group C Inspection (Continued)

INSPECTION	REQT. PARA.	INSP. PARA.
Subgroup IV:		
Electromagnetic Interference	3.6	4.10
Operating Power	3.7.11.1	4.19
Transient spike voltage	3.7.11.5	4.19.1
Beamwidth	3.10.1.2,3.10.2.2	4.17.1.1, 4.17.2.1
Antenna gain	3.10.1.3,3.10.2.3	4.17.1.2, 4.17.2.2
Antenna isolation	3.10.1.5,3.10.2.5	4.17.1.4, 4.17.2.4
Antenna pattern	3.10.1.6,3.10.2.6	4.17.1.5, 4.17.2.5
Signal rejection	3.10.1.8	4.17.1.6

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4.5.3.2.2 Subgroup II - For this subgroup, two units (these units may be the units selected for above Subgroup I), from the first forty (40) production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units every two months shall be selected.

4.5.3.2.3 Subgroup III - For this subgroup, two units from the first forty (40) production units shall be selected for the first Group C inspection. For subsequent Group C inspection, two units shall be selected every four (4) months, except that fungus, acceleration, explosion, dust, and rain shall be selected when one-half of the units on order have been accepted by the Government QA representative.

4.5.3.2.4 Subgroup IV - For this subgroup, two (2) units from the first forty (40) production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two (2) units shall be selected every four months. These units shall not have been subjected to 4.5.3.2.1 through 4.5.3.2.3.

4.5.3.3 Non-Compliance - Actions required relative to group failures shall be as specified in the contract.

4.5.3.4 Re-inspection of Conforming Group C Sample Units - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass Group A inspection.

4.5.4 Group D Inspection - Unless otherwise specified in the contract Group D reliability testing shall be performed by the contractor. This inspection shall consist of the test specified in Table IV and shall be performed on units from lots which have successfully completed the burn-in/debugging and have been subjected to and met Group A and Group B inspections.

4.5.4.1 Sampling for Group D Inspection - A minimum of 3 units and a maximum of 7 units shall be randomly selected by an authorized Government representative for the Reliability Qualification Test (4.6.2) from the First Article samples. For subsequent Group D inspection, a minimum of 3 units and a maximum of 7 units shall be randomly selected by an authorized Government representative for the production reliability tests (4.6.2) from the first production lot after First Article and from every 350 units produced thereafter. The total test time per unit shall be in accordance with the requirements paragraph titled Test Length of MIL-STD-781B.

TABLE IV Group D Inspection

INSPECTION	TEST PARA	REQT PARA
Reliability	4.6.2	3.5

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4.5.4.2 Group D Failures - Actions required relative to Group D failures shall be as specified in the contract.

4.6 Reliability

4.6.1 Burn-In/Debugging and Preconditioning (See 3.11)

4.6.1.1 Burn-In/Debugging (See 3.11.1) - Every unit on order except antennas shall be subjected to, as a minimum, a 48 hour burn-in, the last 24 hours of burn-in shall be failure free. The burn-in shall be conducted under the conditions of 4.6.2.2 and all units shall be exercised and monitored in the same manner as during reliability testing. All failures shall be recorded and analyzed, but shall not be considered relevant for computation of equipment MTBF. The contractor may at his option subject the equipment to an additional Burn-In/Debugging.

*4.6.1.2 Production Run-In (See 3.11.2) - Each RT-1115( )/APN-209(V) and RT-1411( )/APN-209A(V) shall be preconditioned for a period of at least 150 hours prior to being subjected to the burn-in tests. Performance will be checked periodically throughout this preconditioning period, however no formal data will be collected nor required. This preconditioning will be performed using the same temperature and power off-on cycles as the burn-in test but need not include vibration, input voltage variation, or the search-track cycling.

4.6.2 Reliability Testing - Unless otherwise specified in the contract reliability testing shall be performed by the contractor. The contractor shall conduct all reliability tests in accordance with test plan XXII of MIL-STD-781B to determine compliance with 3.5.

4.6.2.1 Conditions Precedent to Reliability Testing - Prior to reliability testing the contractor shall perform the burn-in of 4.6.1 and the thermal and vibration surveys in accordance with paragraphs titled Thermal Survey and Vibration Survey of MIL-STD-781B.

4.6.2.2 Reliability Test Conditions - All reliability testing and burn-in shall be conducted under the conditions of Test Level E of MIL-STD-781B as follows:

(a) The temperature shall be cycled from -54°C to +55°C in accordance with the Alternative Method of Temperature Cycling paragraph of MIL-STD-781B.

(b) The equipment shall be vibrated at 2.2 g ± 10% at any non-resonant frequency between 20 Hz and 60 Hz, measured at the equipment case. The duration of vibration shall be at least 10 minutes of each operating hour.

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(c) The input voltage cycling shall be accomplished as follows: The input voltage shall be maintained at 30.25 VDC for 1 complete temperature cycle, at 27.5 VDC for 1 complete temperature cycle and at 24.75 VDC for 1 complete temperature cycle. This cycling shall be repeated continuously throughout the test.

(d) The equipment duty cycle shall be as follows: The equipment shall operate 1 minute in the search mode and 19 minutes in the track mode during the equipment on time. The equipment shall be on-off cycled once during each complete temperature cycle. The off time shall be 5 minutes.

4.6.2.3 Failure Definition - A failure is defined as the inability of the equipment to perform its required function. This includes any event which causes departure from the performance as required by this specification. The following paragraphs, including subparagraphs of MIL-STD-781B shall apply.

a. Failure Actions

b. Verifying Repair

c. Corrective Action

d. Preventive Maintenance

e. Restoration of Failed Equipment

f. Test Records

4.6.2.4 Parameters to be Measured - As a minimum the following parameters shall be measured and recorded twice daily, once during high temperature stabilization and once during low temperature stabilization. Vibration shall not be avoided during measurement.

Parameter to be Measured	Requirement Paragraph
Altitude Range	3.7.1
Altitude Accuracy	3.7.2.1
System Sensitivity	3.7.5
Self Test	3.7.10
High Altitude and Low Altitude Trip Tests	3.8.4

4.6.2.5 Rework and Retest Provision - Once a reject decision has been reached, no rerun can be made of that test on the same or a new sample from the same lot, unless positive action, consisting of design change, modification, or rework actually relating to observed failures has been incorporated. Any subsequent rerun must start at zero hours operating time.

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4.7 Standard Test Conditions - Unless otherwise specified, inspections shall be performed under the following conditions:

a. Input Voltage: 28.0 ± 2.0 volts DC
b. Warm-Up Time: Not greater than 2 minutes
c. Relative Humidity: Room Ambient
d. Temperature: Room Ambient
e. Atmospheric Pressure: Room Ambient

*4.8 Service Conditions (See 3.4) - Unless otherwise specified, the service condition tests shall be conducted in accordance with MIL-STD-810. Electrical tests and inspections shall be conducted as specified in Table V. No defects shall be allowed. The duty cycle shall be 1.0 minute in search mode and 19 minutes in track mode. These tests apply to the RT-1115( )/APN-209(V), RT-1411( )/APN-209A(V) ID-1917( )/APN-209(V), AS-2595/APN-194(V) and AS-3586/APN-209A(V).

4.8.1 Temperature-Altitude - The unit(s) shall be subjected to the operating and non-operating, altitude and temperature range with supply voltages from 22.5 through 30 VDC as specified in Test Method 504.1, Procedure I, for Category 4 equipment. In addition, the equipment shall also be subjected to the test of Method 504.1, Procedure I, Step 1a for Category I equipment to simulate air transport at an altitude of 40,000 feet. These tests shall be made as indicated to determine compliance with 3.4.1.

4.8.2 Humidity - The unit(s) shall be subjected to the test of Method 507.1, Procedure II. In Step 6 of the procedure, measurements shall be taken during the last 5 hours of the last cycle. These tests shall be made to determine compliance with 3.4.2.

4.8.3 Dust

(1) The antenna shall be subjected to the test of Method 510.1, Procedure I.

(2) The internally mounted unit(s) shall be subjected to the test of Method 510.1, Procedure I with the exception that the air velocity in Steps 1, 2, and 3 shall be 200 ± 100 FPM. Step 3 may proceed immediately after stabilization in Step 2. These tests shall be made to determine compliance with 3.4.3.

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*TABLE V - ENVIRONMENTAL ELECTRICAL TESTS

ENVIRONMENTAL TEST	VIB	TEMP ALT	SHOCK BASIC	ACCEL	HUMID	DUST	EXP ATMOS	RAIN	SALT FOG	LEAK	DIFF PRES
Equipment Reqt Para / Test Para	BEFORE/AFTER DURING	BEFORE/AFTER DURING	BEFORE/AFTER DURING	BEFORE/AFTER DURING	BEFORE/AFTER DURING	BEFORE/AFTER DURING	BEFORE/AFTER DURING	BEFORE/AFTER DURING	BEFORE/AFTER DURING	BEFORE/AFTER DURING	BEFORE/AFTER DURING
Altitude Accuracy 3.2.7.1 4.9.3	X X	X X	X	X	X X	X	X	X	X		
Reliability Signal 3.8.3 4.9.7	X X	X X	X	X	X X	X	X	X	X		
Self Test 3.7.10 4.9.5	X X	X X	X	X	X X	X	X X	X	X		
System Sensitivity 3.7.5 4.9.4	X X	X X	X	X	X X	X	X	X	X		
Display Function 3.9 4.9.8	X X	X X	X	X	X X	X	X	X	X		
Auxiliary Altitude 3.8.2 4.9.6	X X	X X	X	X	X X	X	X	X	X		
Short Term Drift 3.7.2.4 4.9.21	X X	X X	X	X	X X	X	X	X	X		
Duty Cycle of 4.8	X						X	X			
Mechanical and visual inspection and also operation of controls 4.11	To be conducted before and after each service conditions test										
VSWR 3.10.1.4 4.17.1.3 3.10.2.4 4.17.2.3	X	X	X	X	X	X	X	X	X	X	X

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*4.8.4 Vibration - The unit(s) shall be subjected to the test of MIL-STD-810, Method 514.2, Procedure I, Curve M, Figure 514.2-3 except that the upper frequency of vibration shall be limited to 500 Hz. The time schedules shall be in accordance with Table 514.2-III. These tests shall be made to determine compliance with 3.4.4.

4.8.5 Shock

4.8.5.1 Bench Handling - The unit(s) shall be subjected to the test Method 516.2, Procedure V, to determine compliance with 3.4.5.1.

4.8.5.2 Basic - The unit(s) normally mounted shall be subjected to the tests of Method 516.2, Procedure I, with shock pulses in accordance with Figure 516.2-2 for flight vehicle equipment. These tests shall be made to determine compliance with 3.4.5.2.

4.8.5.3 Crash Safety - The unit(s) normally mounted shall be subjected to the tests of Method 516.2, Procedure III, with shock pulses, in accordance with Figure 516.2-2 for flight vehicle equipment. These tests shall be made as indicated, to determine compliance with 3.4.5.3.

4.8.6 Fungus - The units(s) shall be subjected to the test of MIL-STD-810, Method 508.1, Procedure I, except that the cyclic conditions listed should not be used. Instead the conditions of 29°C (84°F) and 95 percent relative humidity should be maintained throughout the test. The unit(s) shall not be specifically cleaned for the fungus test, except for the cleaning it received during or after production. After 28 days, based on visual examination, the unit(s) shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies with mycelial development in areas only in other than critical circuit portions; such as terminal spacing, printed circuit boards, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. The equipment shall fail if it does not meet operational performance criteria or if it shows more than the growth specified above.

4.8.7 Salt Fog - The unit(s) shall be subjected to the test of MIL-STD-810, Method 509.1, Procedure I. Operation shall only take place after washing and drying. In addition to operation there shall be no corrosion of finishes or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish on metal surfaces. There shall also be no clogging or binding of moving mechanical parts. These tests shall be made to determine compliance with 3.4.7.

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4.8.8 Explosive Atmosphere - The unit(s) shall be tested for explosion in accordance with Method 511.1, Procedure I, to determine compliance with 3.4.8.

4.8.9 Acceleration - The unit(s) shall be tested in accordance with Method 513.2, Procedure II of MIL-STD-810. The unit(s) shall be subjected, in turn, in each direction along each of its three orthogonal axes. The test level shall be 6 g for at least one minute. These tests shall be made to determine compliance with 3.4.9.

4.8.10 Rain

(1) The antenna shall be subjected to the test of Method 506.1, Procedure I, of MIL-STD-810.

(2) The IRT unit(s), RT unit(s) and ID unit(s) with connectors covered shall be exposed to a "steady-state" rainfall of 4 ± 1 inches per hour on all surfaces for a total time duration of not less than 2.0 hours. After completion of the rain test, the unit(s) shall be turned on within 30 minutes and shall meet the tests required to determine compliance with 3.4.10.

*4.8.11 Leakage (Immersion) - The Antennas AS-2595/APN-194 (V) and AS-3586/APN-209A(V) shall be tested in accordance with Method 512.1, Procedure III of MIL-STD-810. The absolute pressure of the air above the liquid shall be reduced to 9 inches of mercury and then increased to 10 inches of mercury respectively. The antenna connector shall be connected to a mating connector-cable assembly or shall be capped during this test. The antenna shall meet the requirements of 3.10.1.4 or 3.10.2.4 within 30 minutes of completion of the immersion test to verify compliance with 3.4.11.

*4.8.12 Differential Pressure - Using a suitable test fixture the face of the antenna radome shall be subjected to a differential pressure of one atmosphere for a period of not less than two hours. At the completion of the two hour test period the antenna shall show no mechanical degradation or degradation of VSWR (see 3.10.1.4 or 3.10.2.4).

4.9 Electrical Tests

4.9.1 Electrical Test Conditions (See 3.7)

4.9.1.1 Measurements - Unless otherwise specified, all voltage measurements shall be made with respect to signal ground at J1 pin 4 or J2 pin 11.

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*4.9.1.2 Range Simulation - Range simulation shall b accomplished by: delay of the transmitted Rf pulse from J4 through Rf cable or discrete RF delay components such as Teledyne Model M7432 delay lines, or; by delay of transmitter operation by applying ground to J1 pin 13 and trigger pulses to J1 pin 8 and J1 pin 11. When fixed RF delays are used, attenuation of the fixed delays shall be consistent with that required to perform the tests. The delays for each range shall be as follows:

Simulated Range (feet)	Delay (nsec)
200	406.8
400	813.6
800	1627.2
1000	2034.1
1500	3051.1

When trigger pulses are used to delay transmitter operation, these pulses shall have the following characteristics:

J1 pin 8 (Simulated T-Zero)

Pulse Width	100 ± 25 nanoseconds
Rise Time	20 nanoseconds maximum
Amplitude	4.0 ± 1.0 V referenced to 4.0 ± 1.0 V (negative going pulse). The peak of the pulse shall be within ±0.5 V of ground.
PRF	Synchronous with the pulse at J1 pin 11.

J1 pin 11 (Simulated Video)

Pulse Width	550 ± 100 nanoseconds
Rise Time	20 nanoseconds maximum
Amplitude	4.0 ± 1.0 V referenced to 4.0 ± 1.0 V (negative going pulse). The peak of the pulse shall be witin ±0.5 V of ground.
PRF	18.0 ± 1.5 KHz

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Delay	Up to 850 nanoseconds before and up to 4000 nanoseconds after the pulse generated at J1 pin 8 to an accuracy of ± 2 nanoseconds (as measured from the leading edges).

4.9.1.3 Digital Signal Output Interpretation - Interpretation of the digital output signal shall be accomplished using a Computer Measurement Co. Model 726B counter or equivalent. The digital signal at J2 pins 10 and 12 shall be applied to one section of a 9615 line receiver, the output of which shall be applied to the A input of the counter. The enable gate output at J2 pins 1 and 4 shall be applied to the other section of a 9615 line receiver, the output of which shall be applied to the B input of the counter. With counter functions set to display the number of pulses at input A during the pulse at input B, the counter display is range in feet.

4.9.2 Altitude Range (See 3.7.1)

4.9.2.1 RT-1115( )/APN-209(V) - Energize the unit and after a 2 minute warm-up verify that the digital and analog displays follow the simulated altitudes shown in steps 1 to 5 below when operated with the specified external RF loop attenuation.

Step	External RF Loop Attenuation (dB) (J3 to J4)	Altitude (ft)
1	50 ± 2	0
2	70 ± 2	200
3	90 ± 2	400
4	100 ± 2	1000
5	100 ± 2	1500

4.9.2.2 ID-1917( )/APN-209(V) - Energize the height indicator and verify that the analog and digital displays follows the analog and digital input signals respectively from zero to 1500 feet.

4.9.3 Altitude Accuracy

*4.9.3.1 RT-1115( )/APN-209(V) and RT-1411( )/APN-209A(V) (See 3.7.2.1 and 3.7.2.2) - Energize the unit and after the two minute warm- up verify that the analog and digital readings meet the requirements shown in steps 1 through 5 of Table VI when operated with the specified external RF attenuation.

4.9.3.2 ID-1917( )/APN-209(V) (See 3.7.2.2 and 3.7.2.3) - Energize the height indicator and verify that the analog (pointer) and digital readout meet the limits shown in steps 1 to 6 of Table VII.

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*4.9.4 System Sensitivity - Energize the IRT or RT with the external RF loop attenuation set to 50 ± 2 dB. After a 2 minute warm-up adjust the unit for zero feet indications. Verify that the unit breaks track and re-acquires the signal within the limits in steps 1 and 2 below. Simulate an altitude of 750 to 1500 feet with a total external RF loop attenuation 100 dB. Increase the RF attenuation slowly and verify that the far range sensitivity meets the requirements of step 3 below (see 3.7.5).

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*TABLE VI - Altitude Accuracy RT-1115( )/APN-209(V) and RT-1411( )/APN-209A(V)

Step	RF Attenuation (dB)	Simulated Altitude (feet)	Analog Pointer** (Ref ± feet)	Digital Display** (feet)	Remote Voltage (volts) J2 pin 2	Remote Digital J2 pins 10 to 12
1	50 ± 2	0	2	-0 to 3	-0.021 to +0.021	0 to 3
2	90 ± 2	400	8	380 to 410	-2.695 to -2.905	385 to 415
3	100 ± 2	800	16	770 to 830	-5.411 to -5.789	773 to 827
4	100 ± 2	1000	20	970 to 1030	-6.769 to -7.231	967 to 1033
5	100 ± 2	1500	30	1450 to 1550	-10.164 to -10.836	1452 to 1548

1/ Ref is remote voltage converted to feet at scale factor -7 mv/ft.
** Not applicable to the RT-1411( )/APN-209A(V).

*TABLE VII - Altitude Accuracy ID-1917( )/APN-209(V)

Step	Simulated Remote Analog Signal J1 pin 9 VDC	Simulated Analog/ Digital Altitude	Analog Pointer (feet)	Digital Display (feet)
1	0	0	-2 to +2	0 to 1
2	-1.050	150	147 to 153	149 to 151
3	-2.800	400	390 to 410	390 to 410
4	-5.600	800	784 to 816	790 to 810
5	-7.00	1000	980 to 1020	990 to 1010
6	10.500	1500	1470 to 1530	1490 to 1510

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Step	Simulated Altitude (Ft)	External RF Atten (dB)
1	0	73 dB to 83 dB (attenuation required to break track), 73 to 85 dB at temperature extremes.
2	0	Within 7 dB of break track value.
3	100-1500	Greater than or equal to 119 dB, 117 dB minimum at temperature extremes.

4.9.5 Self Test (See 3.7.10)

4.9.5.1 RT-1115( )/APN-209(V) - Energize the unit with a simulated altitude of zero feet and with 50 ± 2 dB total external RF attenuation. After 2 minutes of operation, depress the push-to-test knob and verify that the analog pointer reads 1000 ± 100 feet, the digital readout reads 1000 ± 100 feet, and the self test confidence signal (J2 pin 13) measures +4.5 ± 1.0 VDC. Verify that system functions similarly when the external push-to-test line (J2 pin 8) is grounded.

4.9.5.2 ID-1917( )/APN-209(V) - Verify that pin J1-5 is connected to pin J1-16 and pin J2-9 is connected to pin J2-20 only when the push- to-test knob is pressed.

*4.9.5.3 RT-1411( )/APN-209A(V) - Energize the unit with a simulated altitude of zero feet and with 50 ± 2 dB total RF loop attenuation. After 2 minutes of operation, ground J2 pin 8 and verify that the remote analog outut (J2 pin 2) is -7 ± 0.7 VDC, the remote digital altitude output (J2 pin 12 with respect to J2 pin 10) provides 1000 ± 100 pulses when the remote enable signal (J2 pin 4 to pin 1) is present and the self test confidence signal (J2 pin 13) measures 4.5 ± 1.0 VDC.

4.9.6 Auxiliary Altitude Output - Energize the IRT or RT with an external RF loop attenuation of 50 ± 2 dB and a simulated zero altitude. Measure the auxiliary altitude output (J2 in 3) and verify that the auxiliary analog voltage meets the following requirements as the simulated altitude and RF loop is varied (see 3.8.2).

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Simulated Altitude (ft)	Total RF Loop Attenuation (dB)	Auxiliary Analog Voltage (VDC)
0	50 ± 2	-.021 to +0.021
150	70 ± 2	-.997 to -1.103
400	90 ± 2	-2.695 to -2.905
800	100 ± 2	-5.411 to -5.789
1000	100 ± 2	-6.796 to -7.231
1500	100 ± 2	-10.164 to -10.836

4.9.7 Reliability Signal - Energize the IRT or RT at zero altitude with a total external RF loop attenuation of 50 ± 2 dB. After a 2 minute warm-up, cause the system to break track and re-acquire the target by increasing and decresing the RF attenuation while measuring the reliability signal at J1 pin 12. Verify that the reliability signal at J1 pin 12 (with respect to ground J1 pin 4) is +4.5 ± 1.0 VDC when the system is in track and is 0.5 ± 0.5 VDC when in the search mode (see 3.8.3).

4.9.8 Display Function (See 3.9)

4.9.8.1 RT-1115( )/APN-209(V) - Energize the IRT at a simulated altitude of 150 feet with 70 ± 2 dB total external RF loop attenuation. After the 2 minute warm-up verify that the display functions follow each of the steps in Table VIII (see 3.9.1).

4.9.8.2 ID-1917 ( )/APN-209(V) - Energize the height indicator and provide the required analog and digital signals to simulate an altitude of 150 feet. Verify that the display functions follow each of the steps in Table IX (see 3.9.2).

4.9.8.3 Dimming - Energize the IRT and/or height indicator as specified in 4.9.8.1 or 4.9.8.2, respectively. By varying the voltage at J1 pin 7 of the IRT or at J1 pin 6 of the height indicator from +28 VDC to zero VDC verify that the digital display varies from full brightness to full off and that the warning lights vary from full brightness to a level visible only with the AN/PVS-5 night vision device (see 3.7.14).

4.9.9 High Altitude and Low Altitude Trip Tests (See 3.8.4)

4.9.9.1 RT-1115( )/APN-209(V) - Energize the IRT at a simulated altitude of 200 ± 20 feet with a total external RF attenuation of 70 ± 2 dB. Verify that the high altitude trip line (J1 pin 2) will sink 80 milliamperes with a voltage on the output line not greater than 1.2 VDC when the high index is set to 150 feet. Similarily, verify that the low altitude trip line (J1 pin 1) will sink 80 milliamperes with a voltage on the output line not greater than 1.2 VDC when the low index is set to 250 feet. Verify that no damage results with the trips deactivated and 50 VDC applied to the output lines in accordance with 3.8.4.1 and 3.8.4.2.

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TABLE VIII - Display Functions RT-1115( )/APN-209(V)

Step	Simulated Altitude (feet)	RF Loop Atten (dB)	HI Set (ft)	LO Set (ft)	PTT Act	Off-On Switch	Analog Pointer (feet)	Digital Readout (feet)	LO Warning Light	HI Warning Light	Flag in View
1	150	70 ± 2	1000	100	No	On	150	150	Off	Off	No
2	150	70 ± 2	130	100	No	On	150	150	Off	On	No
3	150	70 ± 2	1000	300	No	On	150	150	On	Off	No
4	0	70 ± 2	1500	1200	Yes	On	1000	1000	On	Off	No
5	0	70 ± 2	500	300	Yes	On	1000	1000	Off	On	No
6	0	100 ± 2	500	300	No	On	Behind Mask	Blank	Off	Off	Yes
7	0	87 ± 2	500	300	Yes	On	1000	1000	Off	On	No
8	0	87 ± 2	500	Behind Mask	No	Off	NA	Blank	Off	Off	Yes

TABLE IX - Display Functions ID-1917( )/APN-209(V)

Step	Simulated Altitude (feet)	HI Set (ft)	LO Set (ft)	Reliability Signal	Off-On Switch	Analog Pointer (feet)	Digital Readout (feet)	LO Warning Light	HI Warning Light	Flag in View
1	150	300	100	On	On	150	150	Off	Off	No
2	150	130	100	On	On	150	150	Off	On	No
3	150	1000	300	On	On	150	150	On	Off	No
4	2000	500	300	Off	On	Behind Mask	Blank	Off	Off	Yes
5	150	500	Behind Mask	Off	Off	NA	Blank	Off	Off	Yes

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4.9.9.2 ID-1917( )/APN-209(V) - Energize the height indicator and provide the required analog and digital signals to simulate an altitude of 200 ± 20 feet. Verify that the high altitude trip line (J1 pin 8) will sink 80 milliamperes with a voltage on the output pin not greater than 1.2 VDC when the high index is set to 150 feet. Repeat this test at J2 pin 6. Verify that the low altitude trip line (1 pin 7) will sink 80 milliamperes with a voltage on the output pin not greater than 1.2 VDC when the low index is set to 250 feet. Repeat this test at J2 pin 7. Verify that no damage results with the trips deactivated and 50 VDC applied to the output pins in accordance with 3.8.4.1 and 3.8.4.2.

4.9.10 Multiple Indicator Protection - Energize the IRT at a simulated altitude of zero feet with a total external RF loop attenuation of 70 ± 2 dB. After the 2 minute warm-up verify that each of the following lines can be grounded (connected to J2 pin 11) without affecting the dial presentation (see 3.7.8).

a. J2 pin 13 self test confidence signal

b. J2 pin 3 auxiliary analog altitude signal

c. J2 pin 2 remote analog altitude signal

d. J2 pin 12 remote digital altitude

e. J2 pin 10 remote digital altitude return

f. J2 pin 1 remote digital enable pulse return

g. J2 pin 4 remote digital enable pulse

4.9.11 Power Failure Display Functions - Energize the IRT at a simulated altitude of 50 feet with a total external RF loop attenuation of 70 ± 2 dB. Verify that flag comes into view, the digital display is blanked, and neither warning light is on when the primary power line (28 VDC) is disrupted (see 3.9.1.1).

*4.9.12 Altitude Outputs - Energize the IRT or RT at a simulated altitude of 50 feet with a total external RF loop attenuation of 70 ± 2 dB. Cause the system to break track by increasing the RF attenuation and verify that the remote analog signal (J2 pin 2) measures -13 ± 2 VDC, the auxiliary analog signal (J2 pin 3) measures +0.75 ± 0.5 VDC, and the reliability signal (J1 pin 12) measures 0.5 ± 0.5 volts DC (see 3.8.1).

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*4.9.13 Push-to-Test (PTT) Control - Energize the IRT or RT at a simulated altitude of 0 feet with a total RF loop attenuation of 70 ± 2 dB. Connect J2 pin 8 (push-to-test line) to J2 pin 11 (signal ground) and verify that the IRT has switched to the self test mode with the analog and digital displays showing 1000 ± 100 feet and the self test confidence signal (J2 pin 13) switches from 0 ± 0.5 VDC to +4.5 ± 1.0 VDC (see 3.8.5). Verification that the RT is in the self test mode is as specified in 4.9.5.3.

*4.9.14 Output Protection - Energize the IRT or RT at a simulated altitude of 50 feet with a total external RF loop attenuation of 70 ± 2 dB. After a 2 minute warm-up, verify that each of the following outputs/inputs may be grounded (connected to J2 pin 11) or opened without causing damage to the IRT or RT (see 3.7.7):

a. J1 pin 12 reliability signal

b. J1 pin 8 simulated T-zero

c. J1 pin 13 PRF inhibit

d. J1 pin 11 simulated video

e. J1 pin 2 high warning trip**

f. J1 pin 1 low warning trip**

g. J1 pin 4 signal ground

h. J2 pin 13 self test confidence signal

i. J2 pin 3 auxiliary analog altitude signal

j. J2 pin 2 remote analog altitude signal

k. J2 pin 12 remote digital altitude signal

l. J2 pin 10 remote digital altitude return

m. J2 pin 1 remote digital enable pulse return

n. J2 pin 4 remote digital enable pulse

o. J2 pin 11 signal ground

p. J3 receiver cable connector

q. J4 transmitter cable connector

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r. J1 pin 6 blanking pulse

**Not applicable to the RT.

4.9.15 Input Electrical Power (See 3.7.11.2 and 3.7.11.3)

4.9.15.1 RT-1115( )/APN-209(V) - Energize the IRT at a simulated altitude of 50 feet with a total external RF loop attenuation of 70 ± 2 dB. Set the low altitude index to 500 feet and the high altitude index to 600 feet. Verify that input current required by the IRT, the integral lighting current required, and the digital/warning light dimming control current do not exceed the limits shown in Table X as a function of the operating conditions. These tests shall be performed at +30 VDC and +22.5 VDC.

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TABLE X - Input Electrical Power

Altimeter Mode			Integral Lighting		Dim Control		Max Currents (Amps)			
Search	Track	Self Test	5 VAC/DC	OV	28VDC	OV	IRT 28VDC Power In	RI 28VDC Power In	Int Light	Dim Control
X			X		X		1.1	.60	0.5	0.006
X				X	X		1.1	.60		0.006
X			X			X	1.1	.60	0.5	
X				X		X	1.1	.60		
	X		X		X		1.1	.60	0.5	0.006
	X			X	X		1.1	.60		0.006
			X			X	1.1	.60	0.5	
	X			X		X	1.1	.60		
		X	X		X		1.1	.60	0.5	0.006
		X		X	X		1.1	.60		0.006
		X	X			X	1.1	.60	0.5	
		X		X		X	1.1	.60		

Integral Lighting; J1 Pin 10 to Pin 5 for IRT
J1 Pin 17 to Pin 18 for RI

Dim Control; J1 Pin 7 for IRT
J1 Pin 6 for RI

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4.9.15.2 ID-1917( )/APN-209(V) - Energize the height indicator and provide the required analog, digital, reliability and power input signals to simulate the conditions of Table X. Set the low altitude index to 500 feet and the high altitude index to 600 feet. Simulate a 50 foot altitude for a track mode, 1000 foot altitude for Self Test, and out of track conditions for Search. Verify that the integral lighting, dim control, and the 28 VDC power input does not exceed the limits shown in Table X. These tests shall be performed at +30 VDC and +22.5 VDC.

*4.9.15.3 RT-1411( )/APN-209A(V) - Energize the RT at a simulated altitude of 50 feet with a total external RF loop attenuation of 70 ± 2 dB. Verify that the input current does not exceed 0.8 amperes for search, track and self test modes. These tests shall be performed at +30 VDC and +22.5 VDC.

*4.9.16 Electrical Tracking Rate - This test is to be performed prior to the final assembly of the IRT or RT. Energize the Receiver-Transmitter assembly and by monitoring the video pulse and the track gate on an oscilloscope, verify that the system will track a target moving inbound or outbound at 1000 feet per second (see 3.7.4).

*4.9.17 Search Rate - This test is to be performed prior to the final assembly of the IRT or RT. Energize the Receiver-Transmitter assembly in the search mode and verify, by monitoring the internal range voltage, that the search rate (frequency) is 4 ± 0.5 cycles per second (see 3.7.6).

*4.9.18 Altitude Memory - Energize the IRT or RT at a simulated altitude of 50 feet with a total external RF loop attenuation of 70 ± 2 dB. After the two minute warm-up, verify by measuring the time interval from application of ground to the PRF inhibit line (J1 pin 13) to the drop out of the reliability signal (J1 pin 12) that the altitude memory is 1.8 ± 0.5 seconds (see 3.7.12).

*4.9.19 Installation Delay Adjustment - Energize the IRT or RT with a maximum RF cable length totaling 3 feet with an external RF loop attenuation of 50 ± 2 dB, adjust both analog and digital displays for zero foot reading. Simulate an altitude of 100 ± 20 feet with a total external RF loop attenuation of 70 ± 2 dB. Verify that both the digital and analog displays can be adjusted to 70 feet lower than the displayed altitude (see 3.7.9).

*4.9.20 Frequency - Energize the IRT or RT at a simulated altitude of 750 feet or greater with a total external RF loop attenuation of 110 ± 2 dB. After a 2 minute warm-up verify that the transmitter frequency is 4300 ± 10 MHz (see 3.7.3).

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4.9.21 Short Term Drift - Energize the IRT at a simulated altitude of less than 200 feet with a total external RF loop attenuation of 70 ± 2 dB. After the minimum of two minutes warm-up, record the displayed digital reading. Allow the system to continue to operate for an additional 15 minutes and again record the digital readout. The two digital readings shall not vary more than ± 2 feet. This test can alternatively be performed in conjunction with any of the operational tests (see 3.7.2.4).

*4.9.22 Data Lag - This test is to be performed prior to final assembly of the IRT or RT. Energize the Receiver-Transmitter assembly and verify that neither the remote analog signal nor the auxiliary analog signals lags the internal range signal by more than 100 milliseconds when the system is tracking a simulated target moving at a rate of 1000 feet per second inbound or outbound (see 3.7.13).

4.9.23 Integral Lighting (See 3.7.15) - Verify that the IRT and/or height indicator is red internally lighted in accordance with MIL-L- 25467.

4.9.24 Remote Digital Altitude Signal (See 3.8.6) - Energize the IRT or RT with an external RF loop attenuation of 100 ± 2 dB and a simulated altitude of 1500 feet. Monitor the digital signal at J2-P12 on an oscilloscope and verify that the signal meets the requirements of 3.8.6. Verify that the signal at J2-P10 is the complement of the signal at J2-P12.

4.9.25 Remote Enable Pulse (See 3.8.7) - Energize the IRT or RT with an external RF loop attenuation of 100 ± 2 dB and a simulated altitude of 1500 feet. Monitor the remote enable pulse at J2-P4 and the remote digital altitude signal at J2-P12 on an oscilloscope, verify that the requirements of 3.8.7 are met. Verify that the signal at J2-P1 is the complement of the signal at J2-P4.

*4.9.26 Blanking Pulse (See 3.8.8) - Energize the IRT or RT and verify that the blanking pulse has an amplitude of 6.0 ± 0.75 volts, a pulse width of 2.6 ± 0.3 microseconds, a PRF of 18 + 4, -6 KHz, and that the blanking pulse leads the transmitted RF pulse on J4 by 2.0 ± 0.2 microseconds when the blanking pulse is loaded down with a 10,000 ohm ± 5% resistor.

*4.9.27 Zero Altitude Power (See 3.7.16) - This test is to be performed prior to final assembly of the IRT or RT. Energize the Receiver-Transmitter assembly in the forced track mode and manually slew the unit for zero feet altitude. Verify that the maximum peak output power is 27 dBm or less.

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4.10 Electromagnetic Interference - The unit(s) shall be tested, utilizing the measurement techniques of MIL-STD-462. Test RS02 shall be performed in accordance with Notice 3 of MIL-STD-462 except that the current carrying wire shall also be looped around the case. These tests shall be conducted to determine compliance with 3.6.

4.11 Visual and Mechanical - Prior to assembly into the unit(s), printed circuit boards shall be examined for compliance with the equipment drawings. Assemblies, modules and sections shall be examined for compliance with the equipment drawings and MIL-STD-254. All external moveable controls, access covers and the test button shall be operated mechanically. Assembled unit(s) shall be examined and defects classified and recorded in accordance with MIL-STD-252. This inspection shall be performed to determine compliance with 3.3, 3.3.3 and 3.15.

4.12 Interchangeability - The mechanical dimensions and electrical performance of parts shall be physically and electrically interchangeable among the unit(s). Mechanical dimensions and electrical performance required by the gage and equipment drawings shall be measured to determine conformance with physical and functional interchangeability requirements of 3.12. When a listed dimension or performance requirement is not within specified limits, it shall be considered a major defect.

*4.13 Operational Inspection (See 3.13) - Each RT-1115( )/APN-209(V) and RT-1411( )/APN-209A(V) shall be subjected to the tests of 4.9.3.1, 4.9.4, 4.9.5.1, 4.9.5.3 and 4.9.8.1 as applicable. Each ID-1917( )/APN-209(V) shall be subjected to the test of 4.9.8.2. Each AS-2595/APN-194(V) and AS-3586/APN-209A(V) shall be subjected to the VSWR test of 4.17.1.4 and 4.17.2.3 respectively.

4.14 Weight - Unit(s) shall be weighed to determine compliance with 3.3.2.

4.15 Quality Conformance Inspection of Packaging - Packaging shall be inspected in accordance with MIL-P-116, to determine conformance with the packaging requirements of Section 5.

4.16 Rough Handling Test (Packaging) - When rough handling test in accordance with MIL-P-116 is required by the contract, the operational inspection test 4.13 shall be conducted to determine freedom from operational malfunction caused by rough handling.

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4.17 Antenna Tests

*4.17.1 Antenna, AS-2595/APN-194(V) Tests - If specified in the contract, test data on a lot by lot basis from the antenna manufacturer may be furnished to the Government in lieu of performing tests specified in 4.17.1.1 through 4.17.1.7.

*4.17.1.1 Beam-Width - The antenna beamwidth shall be measured using a suitable test chamber to determine compliance with 3.10.1.2. This test may be performed concurrent with 4.17.1.5.

*4.17.1.2 Antenna Gain - The antenna gain shall be measured using a 4.3 GHz RF source and a calibrated receiver to determine compliance with 3.10.1.3. An acceptable method of measuring the gain is to use three antennas, designated antenna A, B and C. Place A as the transmit antenna and B as the receive antenna. Measure the power level at the input to A and at the output of B. Antennas A and B are placed a distance R apart. Repeat with antenna C as the receive antenna and B as the transmit antenna. Repeat with antenna C as the transmit antenna and antenna A as the receive antenna. The gain of each antenna may be determined by solving the following simultaneous equations:

$$g_A + g_B = 20 \log (12.7\ RF)/C - (L_{AO} - L_{Br})$$

$$g_B + g_C = 20 \log (12.7\ RF)/C - (L_{BO} - L_{Cr})$$

$$g_A + g_C = 20 \log (12.7\ RF)/C - (L_{CO} - L_{Ar})$$

where L_{AO}, L_{BO}, L_{CO} = power level in dBm at the input terminal to the transmit antenna: L_{Ar}, L_{Br}, L_{Cr} = power level in dBm at the output terminal of the receive antenna; and g_A, g_B, g_C = dB gain of the antenna at separation R.

*4.17.1.3 V.S.W.R. - The VSWR of the antenna shall be measured at 4.30 GHz, 4.25 GHz, and 4.35 GHz to determine compliance with 3.10.1.4.

*4.17.1.4 Antenna Isolation - Two antennas shall be mounted on the ground plane. The antenna isolation shall be measured using an acceptable method such as a 4.3 GHz source and a calibrated receiver to determine compliance with 3.10.1.5

*4.17.1.5 Antenna Pattern - The E-plane and H-plane antenna patterns shall be measured at 4.3 GHz using a suitable test chamber to determine compliance with 3.10.1.6.

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*4.17.1.6 Signal Rejection - With second harmonic signals applied to the input connector the antenna output shall be measured to determine compliance with 3.10.1.8.

*4.17.1.7 Radome Marking - The radome shall be subjected to a visual inspection to determine compliance with 3.10.1.10.

*4.17.2 Antenna, AS-3586/APN-209A(V) Tests - If specified in the contract, test data on a lot by lot basis from the antenna manufacturer may be furnished to the Government in lieu of performing tests specified in 4.17.2.1 through 4.17.2.5.

*4.17.2.1 Beam-Width - The antenna beamwidth shall be measured using a suitable test chamber to determine compliance with 3.10.2.2.

*4.17.2.2 Gain - The antenna gain shall be measured using a 4.3 GHz RF source and a calibrated receiver to determine compliance with 3.10.2.3.

*4.17.2.3 V.S.W.R. - The VSWR of the antenna shall be measured at 4.30 GHz, 4.25 GHz, and 4.35 GHz to determine compliance with 3.10.2.4.

*4.17.2.4 Antenna Isolation - Two antennas shall be mounted on the ground plane. The antenna isolation shall be measured using an acceptable method such as a 4.3 GHz source and a calibrated receiver to determine compliance with 3.10.2.5.

*4.17.2.5 Antenna Pattern - The E-plane and H-plane antenna patterns shall be measured at 4.3 GHz using a suitable test chamber to determine compliance with 3.10.2.6.

4.18 Compatibility (See 3.1, 3.7.2) - Interconnect the IRT or RT and the remote indicator in accordance with Figure 2 of MIL-I-49077. Verify that the display indications on both units meet the requirements shown on Tables XI and XII. The attenuators and delay lines shall be placed between J3 and J4 as appropriate, in lieu of connecting J3 and J4 to the antennas as shown in Figure 2 of MIL-I-49077.

4.19 Operating Power (See 3.7.11.1) - Verify that equipment performance is as specified herein when operating from power sources having the characteristics of MIL-STD-704, Notice 2, Category B.

4.19.1 Transient Spike Voltage - Energize the IRT or RT at zero altitude with a total external RF loop attenuation of 50 $\pm$ 2 dB. Apply the transient spike to the 28 VDC power lines, power control lines and IRT dim control lines to determine compliance with 3.7.11.5.

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4.20 Power on Control (See 3.7.11.4)

4.20.1 RT-1115( )/APN-209(V) - Energize IRT at zero altitude with a total external RF loop attenuation of 50 ± 2 dB by turning the power on control clockwise. The unit shall come on and the voltage at J2-Pin 5 shall be within 2V of the applied 28V power in voltage at J1-Pin 9. With the power control on the unit fully counterclockwise and no external voltage applied to J2-Pin 5, the unit will be deactivated, and the voltage at J2-Pin 5 shall be 1 ± 1 VDC. When an external voltage of 25 ± 5 VDC is applied to J2-Pin 5 the unit will be energized.

4.20.2 ID-1917( )/APN-209(V) - Energize the height indicator by applying 28V to J1-Pin 1. Verify that the voltage at J2-Pin 11 is within 2 volts of the applied voltage at J1-Pin 1 when the power on control is turned clockwise. Verify that the voltage at J2-Pin 11 is 1 ± 1 VDC when the power on control is fully counterclockwise and no external voltage is applied to J1-Pin 3 or J2-Pin 11.

*4.20.2 RT-1411( )/APN-209A(V) - Energize the RT at zero altitude with a total external RF loop attenuation of 50 ± 2 dB by applying 28 VDC to J1 pin 9 and 25 ± 5 VDC to J2 pin 5. The unit shall come on With no external voltage applied to J2 pin 5 the unit will be deactivated.

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*TABLE XI

COMPATIBILITY SYSTEM ACCURACY REQUIREMENTS

Range (Feet)	Total RF Loop Attenuation (DB)	IRT/RT				RI	
		Pointer**	Digital**	Rem Analog	Aux Analog	Pointer	Digital
0 (cable)	50 ± 2	Ref ±2	-0 to +3	0 ± .021	0 ± .021	Ref ±2	IRT ±1
50 (delay line)	70 ± 2	Ref ±2	45 - 55	-.350 ± .032	-.350 ± .032	Ref ±2	IRT ±1
750 (delay line)	100 ± 2	Ref ±15	720 - 780	-5.25 ± .179	-5.25 ± .179	Ref +15	IRT ±10
1500 (cable)	100 ± 2	Ref ±30	1450 - 1550	-10.500 ± .336	-10.500 ± .336	Ref ±30	IRT ±10
0 (cable) Push to test actuated by IRT button***	100 ± 2	Ref ±20	900 - 1100	-7.00 ± .70	-7.00 ± .70	Ref ±20	IRT ±10
0 (cable) Push to test actuated by RI button	100 ± 2	Ref ±20	900 - 1100	-7.00 ± .70	-7.00 ± .70	Ref ± 20	IRT ±10

Pointer ref value is the REM Analog Voltage equivalent range at -7 mv/ft.

**Displays not applicable to the RT-1411/APN-209A(V).
***When testing with the RT-1411/APN-209A(V) actuate push to test by grounding J2 pin 8.

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*TABLE XII

COMPATIBILITY SYSTEM INDICATION REQUIREMENTS

Range (Feet)	Total RF Loop Attenuation (DB)	IRT**					RI				
		Flag In View	Pointer*	Digital*	Hi Warning Light	Low Warning Light	Flag In View	Pointer*	Digital*	Hi Warning Light	Low Warning Light
Power off	Power off	Yes	N/A	Blank	Off	Off	Yes	N/A	Blank	Off	Off
750 (Delay line)	70 ± 2	No	750	750	Off	On	No	750	750	Off	On
750 (Delay line)	Break track	Yes	Behind Mask	Blank	Off	Off	Yes	Behind Mask	Blank	Off	Off
1500 (Simulated)	100 ± 2	No	1500	1500	On	Off	No	1500	1500	On	Off
2000 (Simulated)	100 ± 2	yes	Behind Mask	Blank	Off	Off	Yes	Behind Mask	Blank	Off	Off

Power controllable from IRT - Yes**

Power controllable from RI - Yes

*Reference only -- no accuracy requirements

**Not applicable when testing with the RT-1411/APN-209A(V)

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5. PACKAGING

5.1 Preservation - Preservation shall be level A, B or commercial, as specified (see 6.2).

5.1.1 Level A

5.1.1.1 Cleaning - Each Altimeter Set, Electronic AN/APN-209( ) (Units Of) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying - Each Altimeter Set, Electronic AN/APN-209( ) (Units of) shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative Application - No preservative required.

5.1.1.4 Unit Packing - Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

*5.1.1.4.1 Antenna AS-2595/APN-194(V), Antenna AS-3586/APN-209A(V) Indicator, Height ID-1917( )/APN-209(V), Receiver-Transmitter RT-1411/APN-209A(V), or Receiver-Transmitter, Height Indicator RT-1115( )/APN-209(V) - Each antenna, indicator height, receiver-transmitter or receiver-transmitter, height indicator shall be individually unit packed method III as follows: Place each item within a bag or wrap fabricated of material conforming to L-P-378. Place the item within a cushioned box conforming to PPP-B-1672, type 1 style A. Close the box in accordance with the box specification.

5.1.2 Level B - Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial Preservation - Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing - Packing shall be level A, B or commercial as specified (see 6.2). Shippping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

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*5.2.1 Level A - Each Altimeter Set, Electronic AN/APN-209( ) or quantity of Antenna AS-2595/APN-194(V), Antenna AS-3586/APN-209A(V), Indicator, Height ID-1917( )/APN-209(V), Receiver-Transmitter RT-1411( )/APN-209A(V) or Receiver-Transmitter, Height Indicator RT-1115( )/APN-209(V), unit packed as specified in 5.1 and bearing the same stock number, shall be packed within a close-fitting box conforming to PPP- B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.1.1 Unitization - Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantites per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film conforming to L-P-378, type IV.

*5.2.2 Level B - Each Altimeter Set, Electronic Set, Electronic AN/APN-209( )(V) or a quantity of Antenna AS-2595/APN-194(V), Antenna AS-3586/APN-209A(V) Indicator, Height ID-1917( )APN-209(V), Receiver-Transmitter RT-1411( )/APN-209A(V) or Receiver-Transmitter, Height Indicator RT-1115( )/APN-209(V), unit packed as specified in 5.1 and bearing the same stock number, shall be packed within a close-fitting box conforming to PPP-B-640, class 2, style E or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will

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be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A or shrink film conforming to L-P-378, type IV. When a single unit is shipped to a single destination no packing is required; the unit conainer shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization - Palletization shall be required when quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of Antenna AS-2595/APN-194(V), Antenna AS-3586/APN-209A(V) Indicator, Height ID-1917( )/APN-209(V), Receiver-Transmitter RT-1411( )/APN-209A(V) or Receiver-Transmitter, Height Indicator RT-1115( )/APN-209(V), unit packed as specified in 5.1 and bearing the same stock number, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV.

5.2.3 Commercial Packing - Packing shall be in accordance with MIL-STD-1188.

5.3 Marking

5.3.1 Military Marking - In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial Marking - In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

6. NOTES

*6.1 Intended Use - The Altimeter Set, Electronic AN/APN-209( )(V) provides a continuous indication of the absolute altitude of an aircraft relative to the surface of the earth or other physical obstructions when installed in fixed or rotory-winged aircraft. The Altimeter Set, Electronic has various configurations as identified in Table XIII.

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TABLE XIII. Altimeter Configurations

Device Nomenclature	Altimeter Set							
	AN/APN-209A(V)			AN/APN-209B(V)				
	(V)1	(V)2	(V)3	(V)1	(V)2	(V)3	(V)4	(V)5
RT-1115A/APN-209(V)	1	1	1	-	-	-	-	-
RT-1115B/APN-209(V)	-	-	-	1	1	-	-	-
RT-1411/APN-209A(V)	-	-	-	-	-	1	1	1
ID-1917/APN-209(V)	-	1	2	-	-	-	-	-
ID-1917A/APN-209(V)	-	-	-	-	1	-	1	-
AS-2595/APN-194(V)	2	2	2	2	2	-	2	2
AS-3586/APN-209A(V)	-	-	-	-	-	2	-	-

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6.2 Ordering Data - Procurement documents should specify the following:

(a) Title, number and date of this specifications and any amendment thereto.

(b) Level A, B or commercial preservation and packing (see section 5).

(c) When rough handling and functional tests are required (see 4.16).

(d) First article units required (see 3.2).

(e) First article test plan and report required (see 4.3.3).

(f) Action required relative to Group C and Group D Failures (see 4.5.4.2 and 4.5.3.3).

(g) Marking and shipping (see 5.3).

(h) Antenna test data required (see 4.17).

(i) Issues of documents (see 2.1).

(j) Government loaned property (see 3.7.14).

*6.3 Location of Operation Inspection - It is desirable that the operational inspection 4.13 be performed at a location that will minimize handling (which might cause damage to the units) after this inspection. It is recommended that the entire lot (including the sample units) be inspected at the packaging site immediately prior to packaging (see 4.13).

6.4 Nomenclature - The parentheses in the nomenclature shall be deleted or replaced by a letter identifying the particular design, for example AN/APN-209W. (Reference MIL-STD-196). The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.5 Changes from Previous Issue - In this specification, an asterisk "*" preceding a paragraph number denotes a paragraph in which changes have been made from the previous issue. This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, regardless of the asterisk notations and their relationship to the last previous issue.

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6.6 System Design, Installation and Application Factors - System design, installation and application factors are provided in MIL-I-49077(CR)

6.7 Definitions -

Remote Enable Pulse Lead Time	Time from the 50% point on the leading edge of the enable gate to the 50% point on the leading edge of the first altitude pulse.
Remote Enable Pulse Lag Time	Time from the 50% point on the trailing edge of the last altitude pulse to the 50% point on the trailing edge of the enable pulse.
Pulse Leading Edge	Negative transition of the altitude signal and enable signal with respect to their return line.

6.8 Environmental - Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.9 Subject Terms (key word) List.

Altimeter
Altimeter Set
Altimeter Set, Electronic AN/APN-209( ) (V)
Altitude
Antenna AS-2595/APN-194(V)
Antenna AS-3586/APN-209( ) (V)
Distribution Statement A
Indicator, Height ID-1917/APN-209( ) (V)
Radar Altimeter
Receiver-Transmitter, Height Indicator, RT-1115( )/APN-209(V)
Receiver-Transmitter RT-1411/APN-209A(V)

Custodian:
Army - CR

Preparing Activity:
Army - CR
Project 5841-A034

Review Activities:
Army -AV, ER

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